Price and Inventory Optimization Report

Maximizing Expected Profit with Joint Pricing and Quantity Decisions under Uncertainty

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1 Summary

This study examines the benefit of integrating pricing and production decisions in a newsvendor setting using historical price–demand data. A simple linear regression of demand on price produced a slope of -1,367.71 and an intercept of 1,924.72, with $R^2 = 0.621$ and RMSE = 149.45, indicating a strong negative relationship and substantial captured variability.

At a fixed price of $p = 1.00$, the optimal production quantity is 628.54 units, yielding an expected profit of 178.99. When price is also optimized, the model selects $p = 0.954$ and $q = 535.29$, achieving an expected profit of 234.43, illustrating the value of jointly optimizing pricing and inventory.

Bootstrap-based sensitivity analysis shows that these decisions are generally stable under sampling variability. A single bootstrap iteration produced a slope of -1,460.08, with optimal price 0.928 and quantity 525.41. Across 100 iterations, the mean optimal price was 0.953 (SD = 0.012) and quantity 538.68 (SD = 30.99), with no infeasible solutions. Scatter and joint plots highlight a negative relationship between price and quantity, while a histogram of expected profits shows a stable distribution centered around $234.

Comparing the optimized model to a standard newsvendor using the historical average price (1.02) and quantity (538.14), which yields profit 84.13, demonstrates a dramatic increase: the integrated policy improves expected profit by 150.30 (179%). These results clearly show that incorporating both pricing and production decisions leads to much better outcomes than relying on a simple fixed-price approach. The model could be extended further by considering more complex demand functions, multi-period inventory planning, or incorporating uncertainty in costs and prices to make the recommendations even more realistic and robust.

2 Introduction

2.1 Production Decisions under Uncertain Demand

Manufacturers must choose production levels before knowing actual customer demand. When extra units cannot be sold, as in the standard newsvendor model, every unit produced above demand becomes a sunk cost. Producing too few units leads to lost sales and missed revenue. The newsvendor model captures this trade-off by selecting a quantity that maximizes expected profit under uncertain demand. However, its basic version assumes fixed prices and ignores practical issues such as disposal fees or rush production, which can cause it to underestimate the real financial risks of poor production choices.

2.2 Model Extensions

In the interest of improved performance, we may want to update the classical framework and include more realistic features. Adding emergency production options, salvage penalties,

or price-sensitive demand links expected profit not only to production quantity but also to pricing decisions and unexpected demand shifts. These extensions often produce linear or quadratic programs that can be solved directly using observed demand data. They also allow firms to study how uncertainty in estimated parameters affects recommended decisions through simulation or bootstrapping.

2.3 Project Objectives

This report uses pricing and demand data to estimate a simple linear demand model, reformulate the extended newsvendor problem for optimization, and identify profit-maximizing price and production levels. It then evaluates how sensitive those recommendations are to sampling noise and model assumptions. The goal is to see whether a complex operational model leads to better pricing and production decisions than the traditional newsvendor model.

3 Methodology

3.1 Simple Linear Regression

We begin by loading the price–demand dataset and fitting a simple linear regression using Scikit-learn to estimate a linear relationship between price and demand. We then compute the in-sample MSE, RMSE, and R^2 to assess how well the model explains variation in demand. The fitted coefficients and residuals from this regression will be used to generate demand scenarios for the optimization steps that follow.

3.2 Linear Program for Optimal Quantity at a Fixed Price

To analyze the fixed-price setting, we set $p = 1$ and construct simulated demand by evaluating the fitted regression at $p = 1$ and adding back the original residuals. This produces a demand vector whose mean equals the predicted demand at $p = 1$ while preserving the day-to-day variability observed in the data.

We then formulate a linear program in Gurobi that maximizes expected profit. Let h_i represent the profit for observation i. Since daily profit is $p \min(q, D_i) - qc$, the auxiliary variables h_i allow us to linearize the $\min(\cdot)$ term:

$$\max_{q,h} \quad \frac{1}{n} \sum_{i=1}^{n} h_i$$

$$\text{s.t.}$$

$$h_i \leq pD_i - qc$$

$$h_i \leq pq - qc$$

This formulation returns the quantity q that maximizes expected profit at the fixed price $p = 1$.

Which we implement in Gurobi almost directly by:

```
...
model.setObjective((1/n) * gp.quicksum(h[i] for i in range(n)), gp.
    GRB.MAXIMIZE)

for i in range(n):
    D_i = demand_scenarios[i]
    model.addConstr(h[i] <= p_fixed * D_i - c_val * q - g_val * u[i]
        - t_val * v[i])
    model.addConstr(h[i] <= p_fixed * q - c_val * q - g_val * u[i] -
        t_val * v[i])
    model.addConstr(u[i] >= D_i - q)
    model.addConstr(v[i] >= q - D_i)
...
```

3.3 Joint Price and Quantity Optimization When Demand Depends on Price

In this extension, price is no longer fixed. Instead, demand is modeled as a linear function of price,

$$D_i = \beta_0 + \beta_1 p + \varepsilon_i,$$

which means that pricing and production decisions must be chosen jointly. Because the revenue term pD_i becomes quadratic in p, the optimization problem is no longer a linear program.

We reformulate the problem in Gurobi as a quadratic program that maximizes expected profit. The decision variables are the price p, the production quantity q, and a set of auxiliary variables h_i that capture the negative of the daily cost for each demand scenario. The expected revenue term is

$$\beta_0 p + \beta_1 p^2$$

since the residuals have mean zero, and the expected negative cost is represented by the average of the h_i variables. The objective function is therefore

$$\max_{p,q,h} \quad \beta_0 p + \beta_1 p^2 + \frac{1}{n} \sum_{i=1}^{n} h_i$$

Daily costs depend on whether production falls short of demand (requiring rushed units at cost g) or exceeds demand (incurring a disposal cost t). To linearize these two cases, each h_i is constrained to be no greater than the negative cost under underproduction and no greater than the negative cost under overproduction:

$$h_i \leq (g - c)q - g(\beta_0 + \beta_1 p + \varepsilon_i)$$

$$h_i \leq -(c + t)q + t(\beta_0 + \beta_1 p + \varepsilon_i).$$

This formulation produces optimal values for both price and production quantity, providing a fully integrated operating policy rather than treating pricing and production as separate decisions.

Which is again implemented in Gurobi, mirroring the equations:

```
...
    avg_h = gp.quicksum(h[i] for i in range(n)) / n
    model.setObjective(revenue_part + avg_h, gp.GRB.MAXIMIZE)

    for i in range(n):
        eps = residuals[i]
        model.addConstr(h[i] <= (g - c) * q - g * (beta_0 + beta_1 *
            p + eps))
        model.addConstr(h[i] <= -(c + t) * q + t * (beta_0 + beta_1
           * p + eps))
...
```

3.4 Bootstrap-Based Sensitivity Analysis of Optimal Price and Quantity

In this section, we assess how sensitive the optimal pricing and quantity decisions are to the specific sample of demand data that was originally observed. Because the regression coefficients and residual distribution are estimated from a finite dataset, small changes in the data can induce changes in the estimated demand curve, and therefore in the optimal solution to the newsvendor optimization problem. To quantify this uncertainty, we apply a nonparametric bootstrap procedure.

Single Bootstrap Iteration

We begin by drawing a bootstrap sample of size n from the original dataset by sampling observations with replacement. This resampled dataset contains repeated observations and omits others, thereby mimicking the sampling variability that would arise if new data were collected from the same underlying demand process.

A new linear regression model is then fit to the bootstrapped data, yielding new coefficient estimates $\beta_0^{(b)}$ and $\beta_1^{(b)}$. Using these estimates, we compute the corresponding residuals

$$\varepsilon_i^{(b)} = y_i^{(b)} - \left(\beta_0^{(b)} + \beta_1^{(b)} p_i^{(b)}\right),$$

These bootstrapped parameters and residuals are then supplied to the nonlinear newsvendor optimization function, `solve_nlp_newsvendor()`, which returns an optimal price $p^{(b)}$, optimal quantity $q^{(b)}$, and expected profit $Z^{(b)}$ under this resampled scenario. Comparing $(p^{(b)}, q^{(b)})$ to the original optimal solution illustrates how a small perturbation of the data can shift the recommended decision.

Repeated Bootstrap

To obtain a full empirical distribution of optimal prices and quantities, we repeat the bootstrap process $B = 100$ times.

Iterations for which the optimization problem does not yield a feasible solution are discarded (though this is technically possible, it is highly unexpected to occur). The retained solutions form empirical distributions for the optimal price, optimal quantity, expected profit, and slope coefficient β_1.

Analysis of Bootstrap Output

Scatter plots of $(p^{(b)}, q^{(b)})$ allow us to visualize how the optimal quantity varies as a function of the optimal price across bootstrap samples. Joint density visualizations highlight whether price and quantity uncertainty are positively or negatively related.

A histogram of $Z^{(b)}$ provides an empirical distribution of expected profits, and the sample mean

$$\bar{Z} = \frac{1}{B} \sum_{b=1}^{B} Z^{(b)}$$

serves as an estimate of expected profitability under parameter uncertainty. Similarly, the sample mean and standard deviation of the bootstrapped prices and the bootstrapped quantities quantify the magnitude of decision variability induced solely by sampling noise in the demand data.

Overall, the bootstrap procedure measures how sensitive the optimal decisions are to the specific dataset used for estimation, and therefore provides a nonparametric assessment of decision robustness in the pricing and quantity optimization problem.

3.5 Comparison of the Optimized Model to the Standard Newsvendor Approach

To evaluate the advantage of the optimized model, we compare it to a traditional newsvendor approach that relies on historical averages.

For the standard model, we assume a fixed price equal to the historical mean and compute the order quantity using the classical critical-fractile formula. Let $\bar{p}$ and $\bar{D}$ denote the historical average price and demand, and σ_D the standard deviation of demand. The order quantity is calculated as

$$q_{\text{standard}} = \bar{D} + z_\alpha \sigma_D,$$

where z_α is the standard normal quantile corresponding to the critical ratio $(\bar{p} - c)/\bar{p}$.

The expected profit of the standard model is then computed using the same cost structure as the optimized model, including production, rush printing, and disposal costs. The optimized model uses the previously described joint price and quantity optimization procedure to determine the optimal price and quantity, and its expected profit is calculated similarly.

This methodology allows a direct comparison between the two approaches in terms of expected profit, optimal order quantity, and pricing decisions, while accounting for variability in demand and the additional costs in the operational setting.

4 Results & Discussion

4.1 Simple Linear Regression

Parameter / Metric	Estimate
Slope (β_1)	-1367.71
Intercept (β_0)	1924.72
Mean Squared Error (MSE)	22,335.89
Root Mean Squared Error (RMSE)	149.45
R^2	0.621

Table 1: Estimated coefficients and in-sample fit metrics for the regression of demand on price.

The regression results show a strong negative relationship between price and demand, with the slope indicating that an increase of one dollar in price reduces demand by roughly 1,368 units. The R^2 of 0.621 suggests the model captures a substantial portion of demand variability, though some residual variation remains. This residual variation is explicitly incorporated when generating stochastic demand scenarios, reflecting the inherent uncertainty in daily demand. The RMSE of 149.45 reflects the typical deviation of observed demand

from the predictions. These estimates and residuals provide the basis for generating realistic demand scenarios for the optimization steps that follow.

4.2 Linear Program for Optimal Quantity at a Fixed Price

Parameter	Value
Price (p)	1.00
Optimal Quantity (q_{opt})	628.54
Expected Profit	178.99

Table 2: Optimal production quantity and expected profit for the fixed-price setting.

At a fixed price of $p = 1$, the linear program identifies an optimal production quantity of approximately 629 units, which maximizes expected profit given the simulated demand scenarios. The expected profit under this strategy is roughly 179.

4.3 Joint Price and Quantity Optimization When Demand Depends on Price

Parameter	Value
Optimal Price (p_{opt})	0.954
Optimal Quantity (q_{opt})	535.29
Expected Profit	234.43

Table 3: Optimal price, production quantity, and expected profit when demand depends on price.

When demand is modeled as a linear function of price, the quadratic program identifies an optimal price of approximately \$0.95 and an optimal production quantity of roughly 535 units. This strategy yields an expected profit of about 234, substantially higher than the fixed-price benchmark. By jointly optimizing price and quantity, the model accounts for the tradeoff between higher prices, which reduce demand, and lower prices, which increase demand but may reduce profit per unit. Including the residuals from the regression captures the day-to-day fluctuations in demand, giving a practical policy that considers both pricing and production together.

4.4 Bootstrap-Based Sensitivity Analysis of Optimal Price and Quantity

Single Bootstrap Iteration

Parameter	Original Value	Bootstrapped Value
Slope (β_1)	-1367.71	-1460.08
Optimal Price (p)	0.954	0.928
Optimal Quantity (q)	535.29	525.41

Table 4: Comparison of original and single-iteration bootstrapped estimates for the slope, optimal price, and quantity.

A single bootstrap iteration slightly changed the estimated slope and the resulting optimal price and quantity. The slope became somewhat steeper, leading to a slightly lower optimal price and quantity. This illustrates that even small changes in the dataset can produce noticeable adjustments in the recommended decisions, highlighting the sensitivity of the optimization to the underlying demand data.

Repeated Bootstrap Analysis

Parameter	Mean	Standard Deviation
Optimal Price (p)	0.953	0.012
Optimal Quantity (q)	538.68	30.99

Table 5: Mean and standard deviation of optimal price and quantity across 100 bootstrap iterations.

Over 100 bootstrap iterations, the optimal price and quantity exhibit modest variability. The mean optimal price is about \$0.95 with a standard deviation of 0.012, while the mean optimal quantity is roughly 539 units with a standard deviation of 31. No infeasible solutions occurred in any of the iterations. These results indicate that the optimized decisions are relatively stable under sampling variability in the demand data, although the quantity recommendation shows more sensitivity than price.

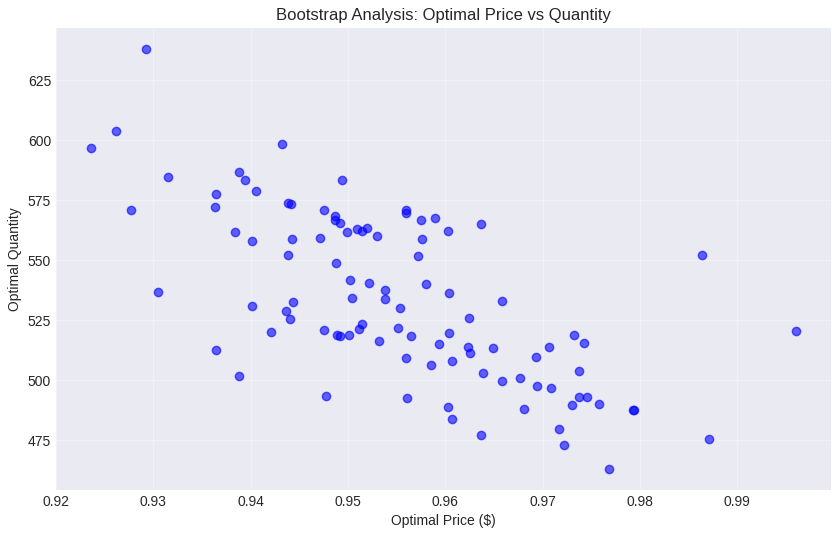

Figure 1: Scatter plot of optimal price vs. quantity across bootstrap iterations.

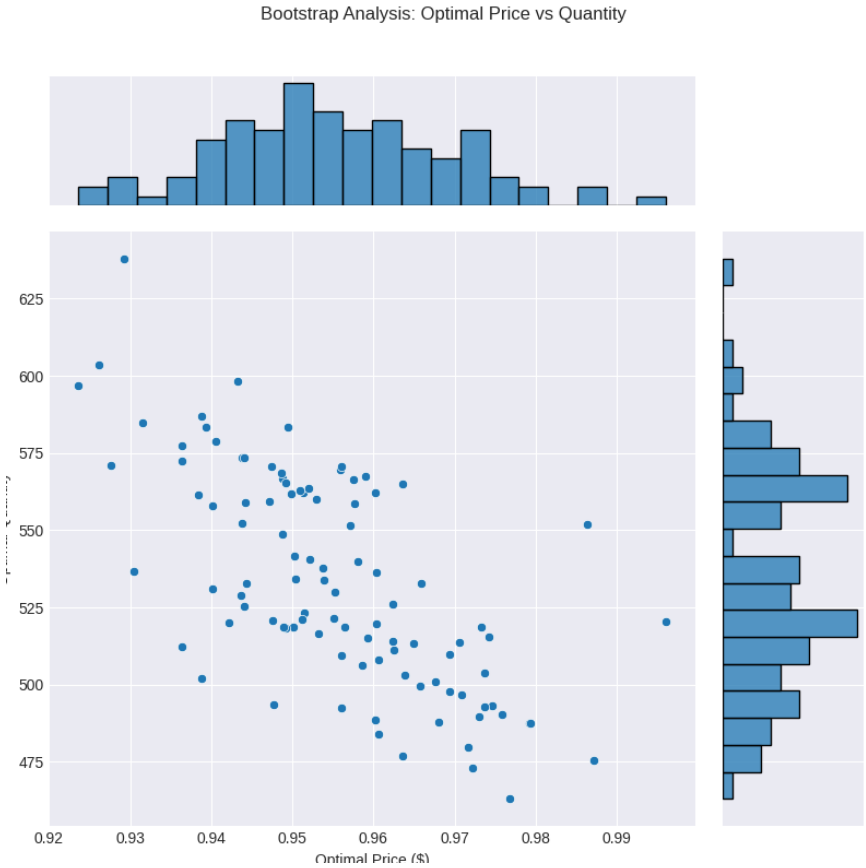

Figure 2: Joint plot of optimal price and quantity, including marginal histograms, across bootstrap iterations.

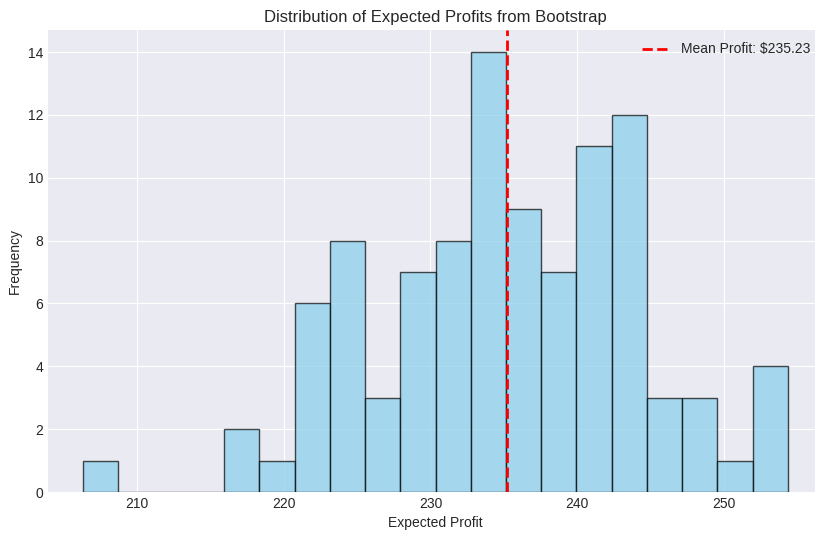

Figure 3: Histogram of expected profits from bootstrap iterations with mean indicated by a red dashed line.

Figures 1 and 2 illustrate how the optimal decision variables change across the 100 bootstrap iterations. The plots reveal a clear negative relationship between the optimal price p and the optimal quantity q. These two variables move in opposite directions. When the data suggests a higher price, the optimization model compensates by recommending a lower order quantity. We can see this in the scatter plot. Iterations with prices near \$0.99 correspond to quantities around 460 units. Conversely, iterations with lower prices near \$0.93 correspond to much higher quantities around 640 units. This happens because the optimization adjusts both variables together to maximize profit based on the specific demand curve estimated in each iteration.

The side charts in Figure 2 highlight a significant difference in stability between the two decisions. The distribution of optimal prices is narrow and tightly clustered. The distribution of optimal quantities is much wider and spans nearly 200 units. This confirms that the inventory decision is much more sensitive to small variations in the data than the pricing decision is. We can be confident in the price, but the optimal order quantity fluctuates heavily depending on the specific data sample.

Figure 3 displays the histogram of expected profits $Z^{(b)}$. The values form a symmetrical shape centered around the mean of \$235.23. The expected profit ranges from a minimum of about \$208 to a maximum of \$255. This distribution does not show any extreme outliers or heavy tails. This indicates that the expected profit is relatively stable. The optimal price and quantity fluctuate significantly as seen in Figure 1. However, the optimization process adjusts both variables simultaneously to offset each other. This balancing act keeps the total profit stable and prevents it from varying as drastically as the individual decision variables.

4.5 Comparison of the Optimized Model to the Standard Newsvendor Approach

Model	Price (p)	Order Quantity (q)	Expected Profit
Standard Newsvendor (Fixed Avg Price)	1.02	538.14	84.13
Optimized Model (Dynamic Price)	0.95	535.29	234.42

Table 6: Comparison of the standard newsvendor approach with the optimized model.

Allowing the model to jointly optimize price and quantity substantially increases expected profit. The optimized price is slightly lower than the historical average, and the optimal quantity is similar, but the integrated pricing and production policy produces a large profit improvement of \$150.30, corresponding to a 179% increase over the standard approach. This shows that accounting for price sensitivity and jointly optimizing price and quantity can dramatically boost profits compared to using a simple fixed-price approach.

5 Conclusion

This analysis demonstrates the value of integrating pricing and production decisions in a newsvendor setting. Fitting a simple linear regression to historical price and demand data allowed us to generate realistic demand scenarios for optimization. Using these scenarios, we first identified the profit-maximizing quantity at a fixed price and then extended the approach to jointly optimize both price and quantity, capturing the effect of price on demand.

Bootstrap-based sensitivity analysis confirmed that the optimized decisions are generally stable under sampling variability, with quantity showing more sensitivity than price. Finally, comparing the optimized model to the standard newsvendor approach revealed a dramatic increase in expected profit when pricing and production are considered together. Overall, these results show that richer operational models can provide much better guidance than simple fixed-price rules. Future work could explore more complex demand functions, multi-period inventory decisions, or incorporating uncertainty in costs and prices to make the model even more realistic and robust.